

MICHAEL ASIEDU ASARE

ORCID: 0009-0003-7318-0348 • mikeasieduasare@gmail.com • www.linkedin.com/in/michael-asiedu-asare •
mikeasieduasare.github.io

RESEARCH INTERESTS

I study how changes in a learned system's inputs, environment, internal state, and computational structure affect the properties that support reliable behavior. My current research focuses on robustness, reliability, and evaluation under distribution shift. I am also interested in AI verification and formal methods, AI security, interpretability, alignment and controllability, and safe and reliable AI deployment. I develop methods to characterize, evaluate, and anticipate failures in AI systems under uncertainty, distribution shift, and changing deployment conditions, with the broader goal of designing and building AI systems that are safe, secure, and effective to deploy.

EDUCATION

BSc Biomedical Engineering (First-Class Honours)

2020 – 2024

Kwame Nkrumah University of Science and Technology (KNUST), Kumasi, Ghana

Final Thesis: Sinkhorn-based Optimal Transport Framework for Breast Cancer Diagnosis (Grade A) | **CGPA: 3.68/4.0**

Selected Coursework: Linear Algebra, Calculus, Probability & Statistics, Signals and Systems, Differential Equations

PUBLICATIONS & MANUSCRIPTS

Yeboah BA, **Asiedu Asare M**, Acquah I, Mensah K, Gbemavor-Assonhe M. Attention-Enhanced Hybrid Bidirectional LSTM and Temporal Convolutional Network for Early Detection of Lung Cancer in Low-Dose CT scans. *Biomedical Engineering and Computational Biology*. 2026. doi: 10.1177/11795972261472208

Asiedu Asare M, Acquah I, Yeboah BA, Owusu E. Resilient Sinkhorn-based Optimal Transport Late Fusion Framework for Breast Cancer Diagnosis. *Cancer Informatics*. 2026. doi: 10.1177/11769351261420789

Owusu E, Acquah I, **Asare MA**, Yeboah BA. LiteFallNet: A lightweight deep learning model for efficient real-time fall detection. *Digital Health*. 2025;11. doi:10.1177/20552076251386698

Manuscripts under Review

Asiedu Asare M, Acquah I. Warning Lead Time for Clinical AI Reliability Under Distribution Shift. Under review at *Artificial Intelligence in Medicine*. ID: AIIM-D-26-06247

Yeboah BA, Acquah I, **Asiedu Asare M**, Owusu E, Opoku-Kwarteng M. Conservative Context-Aware Artifact Management for Free-Living Ambulatory ECG Using Risk-Stratified, Morphology-Aware Evaluation. Under review at *Engineering Applications of Artificial Intelligence*. ID: EAAI-26-30691

RESEARCH EXPERIENCE

Graduate Research Fellow — Biomedical Technologies Lab, KNUST (PI: Dr. Isaac Acquah)

2024 – Present

Lead Researcher — Early Warning of Clinical AI Failure Under Distribution Shift

May 2026 – September 2026

- Introduced Warning Lead Time ($\Delta\alpha$), a metric for how much warning a monitor gives before model performance crosses a prespecified failure threshold. Evaluated it in a 2x2 factorial design (Fourier Domain Adaptation vs. linear blending; chest radiography vs. histopathology) across ResNet18, DenseNet121, and ViT-B/16 (36 runs).
- Showed that robustness depends on the type of shift, not just the architecture. Only 1 of 18 runs failed under FDA, versus 18 of 18 under linear blending. In radiography, the architecture ranking inverted between the two mechanisms: ViT-B/16 was least robust under FDA and most robust under linear blending.

- Tested a critical-slowness precursor borrowed from complex-systems theory. Lag-1 autocorrelation rose before failure in only 10 of 19 failures (53%, $p = 1.00$), no better than chance.
- Under review at *Artificial Intelligence in Medicine*.

Clinical AI Safety Lead — G-MASS

April 2026 – June 2026

- Co-designed an open-source evaluation protocol for medical foundation models in low-resource languages (Twi and Ghanaian English).
- Developed robust evaluation probes testing whether the safety behavior of medical AI assistants holds across both languages.
- Co-developed Safety Degradation Score (SDS), a metric quantifying cross-lingual safety drift in foundation models.
- Released the probe set publicly on Hugging Face (<https://huggingface.co/datasets/BioinstLab/GMASS-probe-set-v1.0>)
- Team placed 3rd out of 104 in the Africa AI Safety Prize.

Researcher — Lightweight Bi-LSTM–TCN for Lung Cancer Diagnosis

June 2025 – November 2025

- Co-developed a hybrid Bi-LSTM–TCN classifier with efficient channel attention and depthwise separable convolutions for three-class (normal/benign/malignant) CT classification.
- Added LIME and SHAP explanations and analyzed the seven test-set errors, including the single missed malignancy.
- Achieved 98.06% accuracy and 99.88% AUC on a held-out test set with 279,561 parameters (3.33 MB), small enough for deployment without specialized hardware.
- Result: Second-author publication, *Biomedical Engineering and Computational Biology* (2026).

Lead Researcher — Optimal Transport for Breast Cancer Diagnostics

May 2025 – August 2025

- Formulated multimodal late fusion as an entropy-regularized optimal transport problem, combining a histopathology classifier (EfficientNetB0, BreakHis) and a clinical-risk model (LightGBM, BCSC) trained on separate, unpaired cohorts.
- Designed a clinically weighted cost matrix that penalizes missed cancers more heavily than false alarms. The learned transport plan let clinical risk upgrade a benign biopsy prediction but rarely downgrade a malignant one.
- Stress-tested under simulated modality dropout: recall stayed at 0.96–1.00 while PR-AUC fell from 0.775 to 0.716 at 50% dropout. Outperformed averaging, max-rule, and logistic-regression fusion on PR-AUC.
- Supported by Google's TensorFlow College Outreach Award to the Biomedical Technologies Lab.
- Result: First-author publication, *Cancer Informatics* (2026).

Researcher — Lightweight Attention Models for Fall Detection

January 2025 – March 2025

- Contributed to a 17,751-parameter (0.312 MB) fall detector that runs on-device in 7.07 ms per sample, so no camera feed or cloud upload is needed.
- Reported the two failure types separately rather than accuracy alone: 1.45% of falls missed, and 2.97% of daily activities falsely flagged on FallAID (97.81% accuracy, 98.55% recall), under instance-level splits.
- Result: Third-author publication, *Digital Health* (2025).

Data Science Researcher (Population Health Analytics)

September 2023 – October 2023

Ghana Statistical Service / Department of Computer Engineering

- Engineered dashboards and statistical summaries to support policy insights for demographic and public health stakeholders.
- Utilized Python-based ETL workflows, EDA, and multi-dimensional visualizations to reveal community-level disparities.

Health Policy Informatics Intern

September 2021 – December 2021

Oforikrom Municipal Health Directorate / Department of Computer Engineering, KNUST

- Authored technical feasibility protocols addressing data flow, system requirements, and infrastructure gaps.
- Co-designed a blueprint for a real-time transactional health information system, transitioning from legacy aggregated reporting.

HONORS, AWARDS, & SCHOLARSHIPS

- Centre for AI Security and Access Africa AI Safety Prize Competition — 3rd place (from 104 applicants, 2026)
- Ingressive for Good Data Science Fellowship (2023)

- Provost's List of Excellent Students, College of Engineering, KNUST (2021 – 2023)
- Ghana Cocoa Board High School Scholarship (2018)

TECHNICAL EXPERTISE

AI & Machine Learning

Deep Learning • Foundation Model Evaluation • Distribution Shift Analysis • Uncertainty Quantification • Multimodal Learning • Representation Learning • Optimal Transport • Dynamical Systems Analysis • Agentic Systems • Explainable AI • Robustness Analysis

Programming & Frameworks

Python • PyTorch • TensorFlow • NumPy • Pandas • Scikit-learn • MATLAB • C/C++ • Git • GitHub • LaTeX • Linux

Research Methods

Evaluation Science • Benchmarking • Statistical Analysis • Scientific Computing • Experiment Design • Open-Source Research Engineering

RESEARCH PRESENTATIONS

Asiedu Asare M, Acquah I. Warning Lead Time for Clinical AI Reliability under Distribution Shift. BioMedEng UK, September 2026.

Asiedu Asare M, Biney JA, Ahenkora AB. Enhanced Mammography Analysis with Personalized Risk Assessment using Explainable Deep Learning. Oral & Poster presentation at KNUST Technology Week, July 2024.

Asiedu Asare M, Biney JA, Ahenkora AB. Optimizing Early Breast Cancer Detection: The Role of AI in Medical Imaging and Analysis. Oral presentation at KNUST BME Symposium, May 2024.

Asiedu Asare M, Musah T. Geospatial Visualization and Analysis of Population Census for Targeted Intervention of Adolescent Pregnancy. Oral presentation at Ghana Statistics Week, October 2023.

Asiedu Asare M. Design of Malaria Surveillance System to Enhance Malaria Elimination in urban Ghana. Oral presentation at Engineering in Society Week, KNUST, March 2022.

TEACHING EXPERIENCE

Teaching Assistant

2024 – 2025

Department of Computer Engineering, KNUST & Biomedical Technologies Lab

- Delivered introductory and intermediate AI modules to 80+ students: AI for Healthcare, ML Foundations, and Responsible AI Principles.
- Designed and taught System Design sessions for 10–20 engineering students focusing on real-world project workflows.
- Facilitated academic writing workshops for 10+ undergraduate students, improving research storytelling, argumentation, and technical communication.

LEADERSHIP & SERVICE

Co-founder & Coordinator, TPL Innovations

2022 – 2025

- Founded a student learning initiative delivering 5+ STEM workshops and peer-learning sessions to 100+ students across KNUST.

General Secretary, Biomedical Engineering Students' Society (BMESS)

2023 – 2024

- Managed administrative operations for 200+ members; organized 10+ academic seminars and 3 industry engagement events.

Vice Chair of Academic Affairs, Engineering Students Senate Council

2023 – 2024

- Helped oversee student policy and governance for 1,000+ engineering students; chaired multiple legislative sessions.

Head, Departmental Debate Club

2022 – 2024

- Led 10+ debaters; coached teams to three consecutive 2nd-place finishes in faculty-level competitions.

Lead Drafter, Ghana Society of Biomedical Engineers Student Chapters Charter

2024 – 2025

- Drafted a national governance framework to guide the establishment of student chapters across 5 universities.

PROFESSIONAL ASSOCIATIONS

Ghana Society of Biomedical Engineers

LANGUAGES

English (Full professional) • French (Intermediate) • Twi (Native) • Chinese (Beginner)